

SAMPLE OR SUGGESTED CURRICULUM ALIGNED TO OUTCOMES BASED EDUCATION FOR BACHELOR OF SCIENCE IN ARCHITECTURE

PROGRAM SPECIFICATIONS

I. Program Description

1.1 Degree Name

The program herein shall be called **BACHELOR OF SCIENCE IN ARCHITECTURE**.

1.2 Nature of Field of Study

The practice of architecture is the act of planning and architectural designing, structural conceptualization, specifying, supervising and giving general administration and responsible direction to the erection, enlargement or alterations of buildings and building environments and architectural design in engineering structures or any part thereof; the scientific, aesthetic, and orderly coordination of all the processes which enter into the production of a complete building or structure, performed through the medium of unbiased preliminary studies of plans, consultations, specifications, conferences, evaluations, investigations, contract documents and oral advice and directions regardless of whether the persons engaged in such practice are residents of the Philippines or have their principal office or place of business in this country or another territory, and regardless whether such persons are performing one or all of these duties, or whether such duties are performed in person or as the directing head of an office or organization performing them.

The practice of architecture includes the provision of professional services in connection with the site, physical planning and the design, construction, enlargement, conservation, renovation, remodeling, restoration or alteration of a building or a group of buildings.

1.3 Program Educational Objectives (PEOs)

The objectives of the Bachelor of Science in Architecture are to develop the following outcomes:

- a. Perform standard competencies in accordance with the scope of the global and local practice of architecture.
- b. Show traits of professionalism, sense of responsibility, equality and patriotism.
- c. Receptiveness to new ideas and knowledge through scientific research;
- d. Direct and focus the thrust of architecture education to the needs and demands of society and its integration into the social, economic, cultural and environmental aspects of nation building.
- e. Instil understanding of the basic philosophy and fundamental principles of the multi-dimensional aspects of architecture, and the direct relationship between man and his environment.

1.4 Specific Professions/careers/occupations for graduates

The graduates of an architecture program may go into the following fields of practice:

- a. Architectural Design
- b. Housing
- c. Physical Planning
- d. Urban Design
- e. Community Architecture
- f. Facility Planning
- g. Construction Technology
- h. Construction Management
- i. Building Administration and Maintenance
- j. Real Estate Development
- k. Education
- l. Research and Development.
- m. Restoration/Conservation

1.5 Allied Fields

The fields of study related to architecture are interior design, landscape architecture, urban design, urban planning, regional planning, environmental planning, housing, real estate development, educational management, business management, project management, construction management and technology, building administration and maintenance, engineering, architectural research, as embodied in RA 9266 including its implementing rules and regulations and the Architects' National code.

2. Program Outcomes

The minimum standards for the BS Architecture program are expressed in the following minimum set of learning outcomes:

2.1 Common to all programs in all types of schools

- a. to keep abreast of the developments in the field of architecture practice. (PQF level 6 descriptor)
- b. the ability to effectively communicate orally and in writing using both English and Filipino
- c. The ability to work effectively and independently in multi-disciplinary and multi-cultural teams. (PQF level 6 descriptor)
- d. A recognition of professional, social, and ethical responsibility

2.2 Common to the discipline

- a. Creation of architectural solutions by applying knowledge in history, theory, planning, building technology and utilities, structural concepts and professional practice. (design, BT, UT, PP, ES/AS, HOA/TOA, PLN)
- b. Use of concepts and principles from specialized fields and allied disciplines into various architectural problems. (LA, IA, HC, PLN, EMP, BU 1-2-3, ES/AS, TOA)
- c. Preparation of contract documents, technical reports and other legal documents used in architectural practice adhering to applicable laws, standards and regulations. (DES, PP 1-2-3, BT 3, BT 1-2-3-4-5, HSN, PLN, ENG3,)
- d. Interpretation and application of relevant laws, codes, charters and standards of architecture and the built environment. (DES, BT 1, HSN, BU 1-2-3, PLN)
- e. Application of research methods to address architectural problems. (RMA, ENG 3)

- f. Use of various information and communication technology (ICT) media for architectural solutions, presentation, and techniques in design and construction. (VT 1-2-3, CADD 1-2)
- g. Acquisition of entrepreneurial and business acumen relevant to architecture practice. (PP3, Project Management (basic knowledge on feasibility, economics/finance, social component, design, construction), PP4 – how to sustain an architectural practice; single proprietorship, partnership, corporate practice, PCN,)
- h. Involvement in the management of the construction works and building administration. (PP, BT, BU, Const. Mgmt., ES/AS)

2.3 Common to a horizontal type as defined in CMO 46 s 2012

- For professional institutions: a service orientation in one's profession
- For colleges: an ability to participate in various types of employment, development activities, and public discourses particularly in response to the needs of the communities one serves
- For universities: an ability to participate in the generation of new knowledge such as pioneering concepts and ideas of site and building designs beyond the regular physical and location boundaries and contexts.

Graduates of State Universities and Colleges must, in addition, have the competencies to support "national, regional and local development plans." (RA 7722)

A PHEI, at its option, may adopt mission-related program outcomes that are not included in the minimum set.

3. Sample Performance Indicators

PROGRAM OUTCOMES	PERFORMANCE INDICATORS
Creation of architectural solutions by applying knowledge in history, theory, planning, building technology and utilities, structural concepts and professional practice.	• Ability to conceptualize design development outputs; Assess the client's needs, opportunities, and constraints • Develop and/or review a program with the client • Develop a vision and goals for the project • Develop or review client's design standards and guidelines • Establish sustainability goals for the project • Define the scope of the pre-design services • Develop or review master plan • Establish requirements of site survey(s) • Review site survey(s) • Review geotechnical and hydrological conditions • Evaluate and compare alternative sites • Perform site analysis • Assess environmental, social, and economic conditions related to project • Document and evaluate existing conditions • Identify requirements of regulatory agencies • Prepare and present submittals for governmental approval • Perform code analyses (e.g., building, energy, accessibility)
Use of concepts and	• Ability to incorporate specialized fields and allied disciplines in the

principles from specialized fields and allied disciplines into various architectural problems.	preparation of contract documents; • Identify requirements of regulatory agencies • Prepare and present submittals for governmental approval Analyze and design basic structural elements and systems Coordinate building systems (e.g., structural, mechanical, electrical, fire safety, security, telecommunications/data) and reconcile systems' conflicts Apply sustainable design principles
Interpretation and application of relevant laws, codes, charters and standards of architecture and the built environment.	• Ability to apply relevant laws, codes, charters and standards of architecture and the built environment in the preparation of contract documents; • Perform code analyses (e.g., building, energy, accessibility) • Review project with code officials • Submit documents to approval agencies and obtain approvals
Application of research methods to address architectural problems.	• Ability to relate results of research in architectural design;
Use of various information and communication technology (ICT) media for architectural solutions, presentation, and techniques in design and construction.	• Ability to use multimedia techniques;
Acquisition of entrepreneurial and business acumen relevant to architecture practice.	• Understand architectural office processes / practices ; • Ability to prepare basic feasibility studies; • Obtain and maintain professional and business licenses • Manage project revenues and expenses • Calculate hourly billing rates • Negotiate and establish fees for basic and additional services and reimbursable expenses • Invoice for services rendered and reimbursable expenses • Develop and manage positive client relationships • Develop leadership skills to enable successful practice • Identify and articulate leadership traits required to maintain a successful and healthy office environment in an architecture firm • Contribute your talents in a community-based organization to improve the quality of life
Preparation of contract documents, technical reports and other legal documents used in architectural practice adhering to applicable laws, standards and regulations.	• Ability to prepare complete sets of contract documents; • Perform code analyses (e.g., building, energy, accessibility) • Review project with code officials • Submit documents to approval agencies and obtain approvals • Prepare specifications based on performance criteria • Research, select, and specify materials • Prepare specifications based on performance criteria • Research, select, and specify materials
Involvement in the management of the construction works and building administration.	• Ability to assist in the construction supervision works; • Perform or review a feasibility study to determine the cost and/or technical advisability of a proposed project • Establish preliminary project scope, budget, and schedule • Prepare and/or evaluate estimates of probable construction costs • Perform value engineering of selected building elements

	Perform life cycle cost analysis of selected building elements Respond to Requests for Information (RFI) Conduct on-site observations Document and communicate status to owner and constructor Resolve constructability issues Issue Architect's Supplemental Instructions (ASI) Process shop drawings and submittals Process Change Orders Review and certify contractor's application for payment Review material test reports Record changes to the contract documents Provide substantial and final completion services Prepare and manage design contracts (owner/architect) Prepare and execute professional services contracts (architect/consultant) Attend, conduct, and record meetings throughout all phases Select, manage, and coordinate consultants Partner with the owner's project delivery team Prepare and manage design team schedule and budget (consultant and staff costs) Obtain client authorization to proceed per contract phases Document project status and progress Monitor project construction costs Prepare owner/contractor agreement Conduct post-occupancy evaluation Identify the project design team members and their required scope of services, roles, and responsibilities (e.g., architects, engineers, specialty consultants) Identify the project delivery team's roles and responsibilities (e.g., owner, architect, contractor, program manager) Identify project delivery method
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CURRICULUM

I. Curriculum Description

- a)** The BS Architecture program has a total of 218 credit units. The program comprises of general education, technical courses (mathematics, natural sciences, basic engineering sciences, professional, allied, and technical elective courses).
- b)** The general education courses are in accordance with the requirements of the CHED Memorandum Order No. 59, s. 1996 - The New General Education Curriculum B (GEC -B) to equip graduates with a basis for critical thinking abilities and values formed from other methods and theories of other disciplines .
- c)** The technical courses comprise of a total of 13 units Mathematics; 9 units of Natural Sciences and 18 units of Basic Engineering Courses.
- d)** The curriculum must have at least 44 professional courses with a total of 115 units, 3 specialization courses with a total of 9 units and a 3 unit correlation course. These courses comprise the basic competencies directed by Republic Act 9266 for licensure in architecture.
- e)** The correlation course must be taken on the final year level.

II. Sample Curriculum

1.1. Program of Study

FIRST YEAR

1ST YEAR – 1ST SEMESTER

Subjects	Units			Hours/Week			Total No. of units	Prerequisite/ (Corequisite)
	Lec	Lab	Studio	Lec	Lab	Studio		
College Algebra	3	0		3	0		3	None
Plane Trigonometry	3	0		3	0		3	None
English 1	3	0		3	0		3	
Pilipino 1	3	0		3	0		3	
Architectural Design 1 - Introduction to Design	1		1	1		3	2	None
Architectural Visual Communications 1- Graphics 1	1		2	1		6	3	None
Architectural Visual Communications 2 - Visual Techniques 1	1		1	1		3	2	None
Theory of Architecture 1	2	0		2	0		2	None
PE 1							2	None
NSTP 1							3	None
Sub-total	17	0	4	17	0	12	26	

1ST YEAR – 2ND SEMESTER

Subjects	Units			Hours/Week			Total No. of units	Prerequisite/ (Corequisite)
	Lec	Lab	Studio	Lec	Lab	Studio		
Analytic Geometry	2	0		2	0	0	2	College Algebra and Plane Trigonometry

Solid Mensuration	2	0		2	0	0	2	College Algebra and Plane Trigonometry
Physics 1	2	1		2	3	0	3	College Algebra and Plane Trigonometry
English 2	3	0		3	0	0	3	
Pilipino 2	3	0		3	0	0	3	
Architectural Design 2 - Creative Design and Fundamentals	1		1	1	0	3	2	Architectural Design 1 and Theory of Architecture 1
Architectural Visual Communications 3- Graphics 2	1		2	1	0	6	3	Architectural Visual Communications 1
Architectural Visual Communications 4 - Visual Techniques 2	1		1	1	0	3	2	Architectural Visual Communications 2
Theory of Architecture 2	3	0		3	0	0	3	Theory of Architecture 1
PE 2							2	
NSTP2							3	
Sub-total	18	1	4	18	3	12	28	

SECOND YEAR

2ND YEAR- 1ST SEMESTER

2.4 Subjects	Units			Hours/Week			Total No. of units	Prerequisite/ (Corequisite)
	Lec	Lab	Studio	Lec	Lab	Studio		
Differential and Integral Calculus	3	0		3	0	0	3	College Algebra and Plane Trigonometry
Physics 2	2	1		2	3	0	3	Physics 1

Science, Environment and Society	3	0		3	0	0	3	None
Architectural Design 3 - Creative Design in Architectural Interiors	1		2	1	0	6	3	Architectural Design 2 and Theory of Architecture 2
Building Technology 1 - Building Materials	3	0		3	0	0	3	Co- requisite Architectural Interiors.
Architectural Visual Communications 5 - Visual Techniques 3	1		1	1	0	3	2	Architectural Visual Communications 4
Architectural Interiors	2	0	1	2	0	3	3	Architecture Design 2 Co- requisite BT -1
History of Architecture 1	3	0		3	0	0	3	None
PE 3							2	
Sub-total	18	1	4	18	3	12	25	

2ND YEAR– 2ND SEMESTER

3 Subjects	Units			Hours/Week			Total No. of units	Prerequisite/ (Corequisite)
	Lec	Lab	Studio	Lec	Lab	Studio		
Statics of Rigid Bodies	3	0		3	0	0	3	Physic 1, Differential and Integral Calculus
Surveying	2	1		2	3	0	3	College Algebra and Plane Trigonometry
General Psychology	3	0		3	0	0	3	
Architectural Design 4- Space Planning 1	1	0	2	1	0	6	3	Architectural Design 3
Building Technology 2 - Construction Drawings in Wood, Steel and Concrete (1-Storey)	2	0	1	2	0	3	3	Building Technology 1,/ (Building Utilities 1)*

Building Utilities 1 - Plumbing and Sanitary Systems	2	1	0	2	3	0	3	Physics 2
Tropical Design	3	0		3	0	0	3	Architectural Design 3
History of Architecture 2	3	0		3	0	0	3	History of Architecture 1
PE 4								
Sub-total	19	2	3	19	6	9	26	

*** Co-requisite course**

“Certificate in Drafting Technology shall be granted upon completion of all prescribed courses from 1st year- 1st semester to 2nd year -2nd semester”

THIRD YEAR

3rd YEAR – 1ST SEMESTER

Subjects	Units			Hours/Week			Total No. of units	Prerequisite/ (Corequisite)
	Lec	Lab	Studio	Lec	Lab	Studio		
Strength of Materials	3	0		3	0	0	3	Statics of Rigid Bodies
Computer-Aided Design & Drafting for Architecture	1	0	2	1	0	6	3	VisCom 5 Graphics 2
Architectural Design 5-Space Planning 2	1	0	3	1	0	9	4	Architectural Design 4
Building Technology 3 - Construction Drawings in Wood, Steel and Concrete (2 storey Res. Structure)	2	0	1	2	0	3	3	Building Technology 2 and Building Utilities 1
Building Utilities 2 (BU 2) – Electrical, Electronics and Mechanical Systems	2	1		2	3	0	3	Physics 2
History of Architecture 3	3	0		3	0	0	3	History of Architecture 2
Professional Practice 1 (Laws Affecting the Practice of Architecture)	3	0		3	0	0	3	None
Sub-total	16	1	6	16	0	18	22	

3rd YEAR – 2nd SEMESTER

Subjects	Units			Hours/Week			Total No. of units	Prerequisite/ (Corequisite)
	Lec	Lab	Studio	Lec	Lab	Studio		
Architectural Design 6- Site Development Planning and Landscaping	1	0	3	1	0	9	4	Architectural Design 5
(Computer-Aided Design & Drafting for Architecture 2) Building Information Modeling	1	0	2	1	0	6	3	Computer- Aided Design & Drafting for Architecture
Theory of Structures	3	0		3	0	0	3	Strength of Materials
Building Technology 4- Specification Writing and Quantity Surveying	2	0	1	2	0	3	3	Building Technology 3
Building Utilities 3 (BU 3) - Acoustics and Lighting Systems	2	1	0	2	3	0	3	Physics 2
History of Architecture 4	2	0		2	0	0	2	History of Architecture 3
Professional Practice 2 (Administering the Regular Services of the Architect)	3	0		3	0	0	3	Professional Practice 1
Planning 1 – Site Planning & Landscape Architecture	3	0		3	0	0	3	Surveying; and Science, Environment and Society
Sub-total	17	1	6	17	3	17	24	

FOURTH YEAR

4TH YEAR – 1ST SEMESTER

Subjects	Units			Hours/Week			Total No. of units	Prerequisite/ (Corequisite)
	Lec	Lab	Studio	Lec	Lab	Studio		
English 3 - Technical Report Writing	3	0	0	3	0	0	3	
Literature 1	3	0	0	3	0	0	3	
Steel and Timber Design	3	0	0	3	0	0	3	Theory of Structures
Architectural Design 7- Community Architecture and Urban Design	1	0	4	1	0	12	5	Architectural Design 6
Building Technology 5 - Alternative Building Construction Systems	2	0	1	2	0	3	3	Building Technology 3
Planning 2 - Fundamentals of Urban Design & Community Architecture	3	0		3	0	0	3	Planning 1
Sub-total	15	0	5	15	0	15	20	

Certificate in (Building Technology) and Utilities shall be granted after completion of all prescribed courses from 1st year - 1st semester to 4th year - 1st semester”

4TH YEAR – 2ND SEMESTER

Subjects	Units			Hours/Week			Total No. of units	Prerequisite/ (Corequisite)
	Lec	Lab	Stu-dio	Lec	Lab	Stu-dio		
Architectural Design 8- Design of Complex Structures	1	0	4	1	0	12	5	Architectural Design 7
Housing	3	0		3	0	0	3	Planning 2 and Professional Practice 1
Planning 3 - Introduction to Urban & Regional Planning	3	0		3	0	0	3	Planning 2
Professional Practice 3 (Global Practice in the 21st Century)	3	0		3	0	0	3	Professional Practice 2
Architectural Structures	5	0		5	0	0	5	Steel and Timber Design
Research Methods for Architecture	3	0		3	0	0	3	Junior Standing, English 3
Specialization 1	3	0		3	0	0	3	Architectural Design 7
Sub-total	21	0	4	21	0	12	25	

“Certificate in CADD shall be granted after completion of all prescribed courses from 1st year- 1st semester to 4th year- 2nd semester”

“Certificate in Basic CADD Proficiency shall be granted after completion of all prescribed courses from 1st year- 1st semester after 3rd year- 3rd year 2nd semester”

FIFTH YEAR

5TH YEAR - 1ST SEMESTER

Subjects	Units			Hours/Week			Total No. of units	Prerequisite/ (Corequisite)
	Lec	Lab	Studio	Lec	Lab	Studio		
Philosophy	3	0		3	0	0	3	
Humanities	3	0		3	0	0	3	
Sociology or Anthropology	3	0		3	0	0	3	
Architectural Design 9 - Thesis Research Writing	1	0	4	1	0	12	5	All courses in 1 st year - 4 th year level
Specialization 2	3	0		3	0	0	3	Architectural Design 8
Architecture Comprehensive Course	3	0		3	0	0	3	All courses in 1 st year - 4 th year level
Sub-total	16	0	4	16	0	12	20	

5TH YEAR - 2ND SEMESTER

Subjects	Units			Hours/Week			Total No. of units	Prerequisite/ (Corequisite)
	Lec	Lab	Studio	Lec	Lab	Studio		
Life and works of Rizal	3	0		3	0	0	3	
Taxation and Agrarian Reform	3	0		3	0	0	3	
Politics and Governance with new Constitution	3	0		3	0	0	3	
Architectural Design 10 - Thesis Research Application	1	0	4	1	0	12	5	Architectural Design 9
Specialization 3	3	0		3	0	0	3	Architectural Design 9
Sub-total	13	0	4	13	0	12	17	

III. Sample Curriculum Map

The minimum standards for the BS Architecture program are expressed in the following minimum set of learning outcomes:

- a. to keep abreast of the developments in the field of architecture practice. (PQF level 6 descriptor)
- b. the ability to effectively communicate orally and in writing using both English and Filipino
- c. The ability to work effectively and independently in multi-disciplinary and multi cultural teams. (PQF level 6 descriptor)
- d. A recognition of professional, social, and ethical responsibility
- e. Creation of architectural solutions by applying knowledge in history, theory, planning, building technology and utilities, structural concepts and professional practice. (design, BT, UT, PP, ES/AS, HOA/TOA, PLN)
- f. Use of concepts and principles from specialized fields and allied disciplines into various architectural problems. (LA, IA, HC, PLN, EMP, BU 1-2-3, ES/AS, TOA)
- g. Preparation of contract documents, technical reports and other legal documents used in architectural practice adhering to applicable laws, standards and regulations. (DES, PP 1-2-3, BT 3, BT 1-2-3-4-5, HSNG, PLN, ENG3,)
- h. Interpretation and application of relevant laws, codes, charters and standards of architecture and the built environment. (DES, BT 1, HSNG, BU 1-2-3, PLN)
- i. Application of research methods to address architectural problems. (RMA, ENG 3)
- j. Use of various information and communication technology (ICT) media for architectural solutions, presentation, and techniques in design and construction. (VT 1-2-3, CADD 1-2)
- k. Acquisition of entrepreneurial and business acumen relevant to architecture practice. (PP3, Project Management (basic knowledge on feasibility, economics/finance, social component, design, construction), PP4 – how to sustain an architectural practice; single proprietorship, partnership, corporate practice, PCN,)
- l. Involvement in the management of the construction works and building administration. (PP, BT, BU, Const. Mgmt., ES/AS)

Legends

Introduced – Students gets introduced to concepts/ principles	I
Practiced – Students practices the competencies with supervision	P
Demonstrated-Student practice the competencies across different settings with minimal supervision	D

PROGRAM CURRICULUM MAP												
List of Required Courses	Program Outcomes Code											
I. TECHNICAL COURSES	a	b	c	d	e	f	g	h	i	j	k	l
Mathematics												
College Algebra												
Advanced Algebra												
Plane & Spherical Trigonometry												

Analytic Geometry												
Solid Mensuration												
Differential and Integral Calculus												
Natural/Physical Sciences												
Physics 1												
Physics 2												
Science, Environment and Society												
Basic Engineering Sciences	a	b	c	d	e	f	g	h	i	j	k	l
Statics of Rigid Bodies												
Strength of Materials												
Theory of Structures						l						
Steel and Timber Design						P						
Architectural Structures						P						
Surveying						P						
Allied Courses	a	b	c	d	e	f	g	h	i	j	k	l
Architectural Visual Communications 1- Graphics 1												
Architectural Visual Communications 2- Visual Techniques 1												
Architectural Visual Communications 3 - Graphics 2												
Architectural Visual Communications 4 - Visual Techniques 2												
Architectural Visual Communications 5 - Visual Techniques 3											l	
Professional Courses	a	b	c	d	e	f	g	h	i	j	k	l
Architectural Design 1 - Introduction to Design	l			l	l			P				
Architectural Design 2- Creative Design Fundamentals	P			P	P	P		P				
Architectural Design 3 – Creative Design in Architectural Interiors	P			P	P	P		P				
Architectural Design 4 - Space Planning 1	P			P	P	P		P				
Architectural Design 5 - Space Planning 2	P			P	P	P		P				

Architectural Design 6 - Site Development Planning and Landscaping	P			P	P	P		P				
Architectural Design 7 - Community Architecture and Urban Design	P			P	P	P		P				
Architectural Design 8 - Design of Complex Structures	P			P	P	P		P				
Architectural Design 9 - Thesis Research Writing	P	D		P	P	P		P				
Architectural Design 10- Thesis Research Application	D			D	D	P		P	P	D		
Theory of Architecture 1					I							
Theory of Architecture 2				P	I							
Architectural Interiors	I				I							
Building Technology 1 – Building Materials												P
Building Technology 2- Construction Drawings in Wood, Steel and Concrete (1 Storey Building)			P	P	P	P	P	P				P
Building Technology 3 - Construction Drawings in Wood, Steel and Concrete (2 Storey Building)			P	P	P	P	P	P				P
Building Technology 4 - Specification Writing and Quantity Surveying		P		P	P		P	P				P
Building Technology 5 - Alternative Building Construction Systems	P					P	P	P				P
Building Utilities 1 (BU 1) - Plumbing and Sanitary Systems					P	P	P	P				P
Building Utilities 2 (BU 2) – Electrical, Electronics and Mechanical Systems					P	P	P	P				P
Building Utilities 3 (BU 3) - Acoustics and Lighting Systems					P	P	P	P				P
History of Architecture 1					P							
History of Architecture 2					P							
History of Architecture 3					P							
History of Architecture 4					P							
Professional Practice 1 (Laws Affecting the Practice of				P	P		I					

Architecture)												
Professional Practice 2 (Administering the Regular Services of the Architecture)			P	P		I	P	P			P	P
Professional Practice 3 (Global Practice for the 21st Century)	P		P	P		P	P	P			P	
Planning 1 - Site Planning and Landscape Architecture				P	P	P	P	P				
Planning 2 - Fundamentals of Urban Design and Community Architecture				P	P	P	P	P				
Planning 3 - Introduction to Urban and Regional Planning				P	P	P		P				
Computer-Aided Design & Drafting for Architecture										P		
Building Information Modelling										P		
Research Methods for Architecture		P		P					I			
Tropical Design					P			P				
Housing				P			P	P				
Architecture Comprehensive Course	a	b	c	d	e	f	g	h	i	j	k	l
Architecture Comprehensive Course					D	D		D				
Specialization Courses	a	b	c	d	e	f	g	h	i	j	k	l
Specialization 1												
Specialization 2												
Specialization 3	P		P									
Non-Technical Courses	a	b	c	d	e	f	g	h	i	j	k	l
Social Science 1,2,3,4				I								
Philosophy												
Humanities												
Literature 1												
English 1,2,3		P										
Pilipino 1,2		P										
Life. Works of Jose Rizal												
P.E. 1,2,3,4												
NSTP 1,2												

IV. Sample Means of Curriculum Delivery

- 10.1 To attain the program outcomes, the school/college of architecture shall ensure that the faculty follow a program or set of activities based on the approved course syllabi.
- 10.2 As a means of curriculum delivery, teachings methods and techniques must be designed and implemented to provide a quality learning environment which nurtures the creativity and artistic sensibilities while encouraging students to be responsive to social development and environmental issues.
- 10.3 To improve curriculum delivery, it shall be the policy of the Architectural institutions to support and encourage the following:
 - a. Field visitations to any appropriate on-going construction site and architectural and heritage tours.
 - b. Participation of students in design competitions, relevant seminars, culture and educational activities outside the school for the enhancement of their skills.
 - c. Invitation of leading practicing architects and allied professionals to conduct informative lectures on current architectural trends and engineering technologies.
 - d. Participation in the design of community-based projects including public buildings and sites.
 - e. Other teaching and learning activities which shall enhance student\ competence and proficiency.

V. SAMPLE SYLLBUS

COURSE SYLLABUS

Course Code	AAA 222	
Course Title	PLANNING 2 – FUNDAMENTALS OF URBAN DESIGN & COMMUNITY ARCHITECTURE	
Credit Units	Lecture	3 UNITS
	Laboratory / Studio	(none)
Pre-Requte(s)	Couse Code	AAA 111
	Course Title	PLANNING 1 – SITE PLANNING & LANDSCAPE ARCHITECTURE
Course Description:		
Spatial Order, Socio –Cultural expression in the design of the exterior environment in neighborhoods, communities, towns & cities		
Course Outcomes		
Upon completion of the course, the learner will be able to:		
1. acquire a sense of spatial order, scale, culture and history in handling urban design and community architecture problems; 2. develop awareness on the need for socio-cultural expression and communication in the design of specific place in towns and cities; 3. have a working knowledge of organization and behavior in the design of specific towns and cities. 4. have an awareness on the importance of community involvement/participation and co-design techniques in urban/community architecture. to design the built environment in the context of ecological balance, sustainable development and conservation of cultural and historical heritage.		

COURSE CONTENT					
Week	Hours	Learning Outcomes	Topics	Teaching Learning Activities	Assessment Tasks
1 st	3	At the end of the lesson, the learner will be able to: 1. <u>know</u> the Class Policy of the Class to be followed 2. <u>appreciate</u> Urban Design & Community Architecture	1. Personality Introduction/ Class Policy 2. Introduction to Urban Design & Community Architecture	1.Lecture 2.Class Discussion	1.Class Diagnostic Assessment
2 nd -3 rd	4.5	1. <u>Use</u> socio-cultural elements to urban design 2. <u>Relate</u> historical background to Community Architecture	1. Socio-Cultural Basis of Design of Communities 2. Historical Background: Aesthetics/ Community Architecture	1.Lecture 2.Class Discussion	1. Quiz
3 rd	1.5	1.Distinguish & use Community Identity & means of Orientation	1. Orientation & Identity in Community Architecture	1.Lecture 2.Class Discussion	1. Quiz
4 th -5 th	4	1.Identify & design Sense of Place & Time	1.Creating and Identifying the Sense of Place & Sense of Time	1.Lecture 2.Class Discussion	1. Reports

COURSE CONTENT					
Week	Hours	Learning Outcomes	Topics	Teaching Learning Activities	Assessment Tasks
5 th -6 th	4	1.distinguish & apply signs & symbols in Urban Design 2.Categorize & apply the Elements of Urban Design	1.Signs and Symbols in Urban Design 2.Elements of Urban Design	1. Lecture 2. PPT Presentation 3. Class Discussions 4. Field work	1. Quiz 2. Drawing Exercises 3. Reports
6 th	1	PRELIM EXAMINATION			

COURSE CONTENT					
Week	Hours	Learning Outcomes	Topics	Teaching Learning Activities	Assessment Tasks
7 th -8 th	4	At the end of the lesson, the learner will be able to: 1.Distinguish & compare the different images of cities 2.Categories environmental responses to urban design	1.The Image of the City 2.Responsive Environment	1. Lecture 2. PPT Presentation 3.Class Discussions	1. Quiz 2. Drawing Exercises
8 th -9 th	5	1.Comprehend Urban Spaces 2.Analyze the system used in urban design 3.Recognize & distinguish the different urban patterns 4.Illustrate & make a presentation of existing urban places	1.Space in Urban Design, Urban Aesthetics, Urban Pattern 2.Documenting the City: The system of design and process of presentation	1. Lecture 2. PPT Presentation 3.Class Discussions	1. Reports 2. Drawing Exercises 3. Quiz

10 th -11 th	4	1. recognize, relate & apply theories, rules & processes in urban design	1.Theories, Rules, and Process in Urban Design	1. Lecture	1.Assignments
11 th -12 th	4	1. Identify & make laws (other legal considerations) as framework in designing urban areas	1. Legal Considerations in Urban Design	1. Lecture	1.Assignments 2.Quiz for (10 th -12 th Wks)
12 th	1	MIDTERM EXAMINATION			

COURSE CONTENT					
Week	Hours	Learning Outcomes	Topics	Teaching Learning Activities	Assessment Tasks
13 th -15 th	7	At the end of the lesson, the learner will be able to: 1. Recognize, classify, & compare Special Uses in a developed environment	1.Overview of Special Uses: a. Cross-Roads Communities b. Agricultural Settlements c. Industrial Estates d. Military Camps e. Commercial Complexes f. Transportation Hubs g. Recreational Centers h. Mining Settlements i. Retirement Communities	1. Lecture 2. PPT Presentation 3.Class Discussions 4.Field Work	1. Reports 2. Quiz

15 th -17 th	7	1. Identify the various Design Requirements of Specific Places 2. Apply these Requirements in urban design	1. Design Requirements of Specific Places in Towns & Cities. a. Cluster Housing and Planned Unit Development (PUD) b. Areas for Priority Development (APDs) / Mixed Use Development & com'l ctrs c. Industrial parks and districts d. Planning Educational Campuses e. Government Center & the Plaza Complex f. Coastal/Lakeside Community Planning g. Resort Community Design h. Parks and Open Spaces, Recreational Areas	1. Lecture 2. PPT Presentation 3. Class Discussions 4. Research	1. Assignments 2. Quiz
17 th -18 th	3	1. Formulate a simple design for an urban community 2. Adhere to Urban Design as a Public Policy	1. Design as a Public Policy	1. Lecture 2. Final Class Project	1. Project presentation
18 th	1	FINAL EXAMINATION			

Course References	
Textbook	(none)
Other References	1. Urban Design: The American Experience – Lang, Jon 1994 2. The Image of the City – Lynch, Kevin 3. Communities within Cities – Davies, W.K.D. / Herbert, D.T. 4. Extension Space in Architecture -Ashihara 5. Design of Cities – Bacon, Edmond 6. Streets and Squares, The Green Dimension, Designing the City (1999) (4 vol.) – Burton, Hugh 7. Color in Architecture: design Methods for Building Interiors and Urban Spaces – Linton, H. 8. Urban Landscape Design - Eckbo, Garrett
Grading System	$\text{TPG/TMG/TFG} = (\text{PE} \times 35\%) + (\text{Q} \times 20\%) + (\text{Pr} \times 20\%) + (\text{As} \times 10\%) + (\text{ACP} \times 15\%)$ $\text{MG/FG} = (\text{TPG} \times 1/3) + (\text{TMG} \times 2/3)$ Legend: 1. PE - Periodic Exam 2. Q – Quizzes 3. Pr – Projects 4. As - Assignments 5. ACP - Average Class Participation 6. TPG/TMG/TFG – Tentative Prelim Grade/ Tentative Midterm Grade/ Tentative Final Grade 7. MG/FG – Midterm Grade/ Final Grade

Course Requirement/s	1. Quizzes 2. Periodic Exams 3. Class Discussions 4. Assignments 5. Drawing Plates 6. Oral Reports
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PREPARED BY:

REVIEWED BY:

<<Faculty Name>>

<<Faculty Name>>

<<Name of industry Practitioner >>

<<Faculty Name>>

<<Faculty Name>>

VERIFIED BY:

RECOMMENDING APPROVAL:

APPROVED BY:

<<Name of Department Chair>>

<<Name of Associate Dean>>

<<Name of Associate Dean>>

Department Chair, <<Name of Department> >

Associate Dean, <<College Initials>>

Dean, <<College Initials>>

COLLEGE OF ARCHITECTURE

OBE Course Syllabus

COURSE TITLE	ARCHITECTURAL DESIGN 7	COURSE CODE	D7
COURSE DESCRIPTION	This course deals with the design problems giving emphasis on the importance of solving complex architectural situations, conditions and the perceptual sensitivities of man.		
NO. OF LECTURE HOURS PER WEEK	Lecture: One (1) Design Studio: Nine (9)	CREDIT UNITS	Five (5)
PRE-REQUISITE(S)	SUBJECT CODE	DESCRIPTIVE TITLE	
COURSE OBJECTIVES	D6	ARCHITECTURAL DESIGN 6	
This course aims: 1. To design and present more complex architectural form focusing on the development of architectural planning and design through proper site analysis including site orientation; 2. To apply the most relevant building laws and regulations in architectural presentations; 3. To identify and translate architectural design solutions in relation to its build environment; 4. To present an architectural design solution that considers the importance of the environment and ecological balance; 5. To utilize the methods of research in analyzing and identifying the importance of architecture in terms of site planning and development.			
VALUE AIMS			
The value and importance id applying lessons learned during the previous program on space manipulation through the creation of			

correct space relations for the purpose of comfortable and universal architecture.

Course Intended Learning Outcomes (CILOs)	Teaching Learning Activities (TLAs)	Assessment Tasks (ATs)
At the end of the course the student should be able to: LO1. Design complex architectural forms and choose an appropriate site through proper site analysis including site orientation. LO2. Design complex architectural forms focusing on structural concepts. LO3. Demonstrate the application of building laws and regulations to architectural planning and design. LO4. Develop architectural design solutions in relation to its ecology and build environment. LO5. Employ conceptual design and site planning analysis. LO6. Develop students' ability to utilize the method of research in data gathering for site selection and architectural planning and design solutions.	-Lecture/Discussion -E-support Teaching & Learning - Esquisse -Drawing Exercises -Research -Working Drawings -Oral Presentation	-Individual / Group Project Research / Assignment -Esquisses -Major Plates / Working Drawings Production & Presentation -Major Examination

COURSE PROGRAM

Module	Week	Course Content	Connected LO's	Outcomes-Based Assessment (OBA)	Teaching and Learning Activities (TLA)	Wright to Module to ILO
1.0	1	Course Introduction: -Brief Introduction to Creative Design -Class Requirements -Class Diagnostic Assessment				
2.0	2-6	Emphasis on Building Structures and Utilities -Major Plate No. 1	LO1, LO4, LO5, LO6	Major Plate 1 & 2; Preliminary Exam	-Lecture/Discussion -E-Support Teaching and Learning -Esquisse -Drawing Exercises -Research -Working Drawings -Oral Presentation	20
3.0	7-11	Laws and Structural Concepts -Major Plate No. 2	LO1, LO2, LO3, LO6	Major Plate 1 & 2; Preliminary Exam	-Lecture/Discussion -E-Support Teaching and Learning -Esquisse -Drawing Exercises -Research -Working Drawings -Oral Presentation	20

4.0	9		Preliminary Examination			15
5.0	12-16	Ecological Planning and Design with Nature -Major Plate No. 3	LO1, LO2, LO3, LO4, LO5, LO6,		-Lecture/Discussion -E-Support Teaching and Learning -Esquisse -Drawing Exercises -Research -Working Drawings -Oral Presentation	30
6.0	17-18	Final Evaluation	Final Examination			15

COURSE REQUIREMENT		
Requirements	Grade Percentage Distribution	Final Grade Computation
-Individual / Group Project Research / Assignment -Esquisses -Preliminary & Final Examination	30%	$\text{Final Grade} = 0.30 [(E/R+PE+FE)/3] + 0.20 (MP1) + 0.20 (MP2) + 0.30 (MP3)$ Passing Grade = 75% and above
Course Work: -Major Plate No. 1 -Major Plate No. 2 -Major Plate No. 3	20% 20% 30%	

Where:
MP=Major Plate
E/R=Esquisses/Research Works
PE=Prelim Exam

		FE=Final Exam
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GRADING SYSTEM			REFERENCES
Grade	Equivalent Grade	Description	-Building Types: Time Savers, Mc-Graw Hill -Architectural Graphics Standards, Ramsey, C., 2000 -The Elements of Styles, Stephen Calloway, Mitchell Beazley Publications, 1992
96-100	1.0	Excellent	
94-95	1.25	Very Good	
92-93	1.50	Very Good	
89-91	1.75	Good	
87-88	2.0	Good	
84-86	2.25	Satisfactory	
82-83	2.5	Satisfactory	
79-81	2.75	Fair	
75-78	3.0	Pass	
Below 75	5.0	Fail	
	WF	Withdrew w/o Permission	

	WP	Withdrew w/ Permission	
	FA	Failure due to absences	
Remarks	Maximum Allowable Absences = 10		

Prepared by:	Verified by:	Approved by:
Faculty Member's Printed Name and Signature	Course Coordinator	Dean, College of Architecture
Date:	Date:	Date:

ARCHITECTURAL DESIGN MAJOR PLATE GRADING SHEET

Name:	Yr. & Section:	Adviser:
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Major Plate Title:	Site:
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PROJECT ISSUES	LESS EVIDENT (50-74%)	FAIRLY EDVIDENT (75-83)	VERY EVIDENT (84-91)	EXCEPTIONALLY EVIDENT (92-100)
<i>ACCEPTABILITY AND SUCCESSFUL TRANSLATION OF THE DESIGN CONCEPT 10%</i>	Work did not reflect any design concept/ character; the work had no conceptual basis	Design concept was expressed, although its relationship to the Design Philosophy and its translation had obvious inconsistencies	Design concept was clearly related to the Design Philosophy, Explained and translated in the work, with very minor inconsistencies	Design concept was very clearly related to the Design Philosophy, well-explained and successfully translated in the work
ENCIRCLE THE APPROPRIATE GRADE ----- >	5 5.5 6 6.5 7	7.5 8	8.5 9	9.5 10
<i>SOUNDNESS OF THE SITE DE'VT PLAN: TRAFFIC CIRCULATION, SEGREGATION, ZONING OF AREAS,</i>	Work did not show any logical planning approach or strategy	There was an attempt to put the site in order but a number of incoherent	The site was well-planned, site constraints were solved, maximized the site potentials but	The planning solution demonstrated an exemplary example of good site planning in all

LANDSCAPING 15%		solutions were committed	have some minor flaws	aspects
ENCIRCLE THE APPROPRIATE GRADE ----- >	7.5 8.0 8.5 9.0 9.5 10 10.5 11	11.5 12 12.5	13 13.5 14	14.5 15
SOUNDNESS & CREATIVITY OF THE FLOOR PLANS: IRCULATION, SEGREGATION, ZONING OF AREAS 35%	The work showed poor and unacceptable layout of floor plans	The work showed an attempt to design; spaces were logically acceptable but had violation of design standards, bldg code and other related national / local laws	The work was well-designed, coherent, followed the standards and codes but was observed to have some very minor design faults	The work was very impressive, almost faultless, exhibited fresh ideas in designing and had consistently followed the codes and standards
ENCIRCLE THE APPROPRIATE GRADE ----- >	17 18 19 20 21 22 23 24 25	26 26.5 27 27.5 28 28.5 29	30 30.5 31 31.5 32	32.5 33 33.5 34 34.5 35
INNOVATIVE AND EFFECTIVE INTERPRETATION OF ELEVATIONS AND SECTIOND BASED ON THE DESIGN CONCEPT 20%	The work showed poor and inconsistent interpretation of elevations and sections	The work showed an attempt to design the vertical aspect; logically acceptable but had inconsistencies with the floor plans	The work was well-designed, coherent, but was observed to have some very minor design faults	The work was very impressive, almost faultless, exhibited fresh ideas in designing and had consistently followed the codes and standards
ENCIRCLE THE APPROPRIATE GRADE ----- >	10 11 12 13 14	15 15.5 16 16.5	17 17.5 18	18.5 19 19.5 20
COMPETENT TRANSLATION OF EXTERIOR AND INTERIOR PERSPECTIVES 10%	The work showed poor and inconsistent interpretation of exterior and interior perspectives	The work showed an attempt to interpret the design in 3-dimensional form but had inconsistencies with the floor plans, elevations and sections	The work was well-designed, coherent, but was observed to have some very minor design faults	The work was very impressive, almost faultless, exhibited fresh ideas in designing and rendering 3-dimensional images

ENCIRCLE THE APPROPRIATE GRADE ----- >	5 5.5 6 6.5 7	7.5 8	8.5 9	9.5 10
OVERALL CORRECTNESS OF DRAWINGS AND CREATIVITY OF PRESENTATION 10%	The work did not observe drafting standards, untidy, and did not meet basic acceptable presentation	The work met the minimum acceptable design presentation, but had a number of drafting errors	The work was commendable, neat, impressive but had some minor drafting errors	The presentation was very commendable, well-presented, very neat and had no drafting errors
ENCIRCLE THE APPROPRIATE GRADE ----- >	5 5.5 6 6.5 7	7.5 8	8.5 9	9.5 10

Adviser's Comments:

PROF'S SIGNATURE

SAMPLE COURSE SYLLABUS

1. **Course Code:**
2. **Course Title:** Building Information Modeling 2
3. **Pre-requisite:** Computer-Aided Design and Drafting for Architecture 2
4. **Co-requisite:** None
5. **Credit:** 1 unit / 2 units
6. **Course Description:** An advanced building information modeling for computer aided architectural rendering, modeling and animation using current software.
7. **Student Outcomes and Relationship to Program Educational Objectives**

Student Outcomes		Program Educational Objectives				
		1	2	3	4	5
(a)	to keep abreast of the developments in the field of architecture practice	✓	✓		✓	✓
(b)	the ability to effectively communicate orally and in writing using both English and Filipino	✓	✓		✓	✓
(c)	the ability to work effectively and independently in multi-disciplinary and multi-cultural teams.	✓	✓		✓	✓
(d)	A recognition of professional, social, and ethical responsibility	✓	✓	✓	✓	✓
(e)	Creation of architectural solutions by applying knowledge in history, theory, planning, building technology and utilities, structural concepts and professional practice.	✓	✓			✓

(f)	Use of concepts and principles from specialized fields and allied disciplines into various architectural problems.	✓	✓		✓	✓
(g)	Preparation of contract documents, technical reports and other legal documents used in architectural practice adhering to applicable laws, standards and regulations	✓	✓		✓	✓
(h)	Interpretation and application of relevant laws, codes, charters and standards of architecture and the built environment.	✓	✓		✓	✓
(i)	Application of research methods to address architectural problems.	✓	✓	✓	✓	✓
(j)	Use of various information and communication technology (ICT) media for architectural solutions, presentation, and techniques in design and construction.	✓	✓		✓	✓
(k)	Acquisition of entrepreneurial and business acumen relevant to architecture practice.	✓	✓		✓	✓
(l)	Involvement in the management of the construction works and building administration	✓	✓		✓	✓

8. Course Outcomes (COs) and Relationship to Student Outcomes

Course Outcomes After completing the course, the student must be able to:	Student Outcomes*											
	a	b	c	d	e	f	g	h	i	j	k	l
1. apply building information modeling (BIM) techniques to complete solid architectural projects			P				P			P		
2. Prepare computer-aided architectural	P									P		

perspective renderings												
3. Prepare computer-aided walkthroughs and animations.										P		

* Level: I- Introduced, P- Practiced, D- Demonstrated

9. Course Coverage

Week	TOPICS	TLA	AT	COURSE OUTCOMES
1	Mission and Vision of Mapua Institute of Technology Orientation and Introduction to the Course Discussion on COs, TLAs, and ATs of the course Overview on student-centered learning and eclectic approaches to be used in the course Introduction to Building Information Modeling	Lecture / Discussion demonstration / application	PL 100 – Text	CO 1
2	User Interface	lecture/ open discussion/ demonstration	PL100 Plan	CO 1

Week	TOPICS	TLA	AT	COURSE OUTCOMES
3	Conceptual Design using Massing	lecture/ demonstration / application	PL100 Elevations Sections	CO 1
4	Utilizing Components	lecture/ open discussion / application	PL100 – Perspective Walkthrough and Rendering	CO 2/ CO3
5	Detailing	lecture/ open discussion / application	PL200 – Plan Residential	CO 1
6	Rendering	lecture/ open discussion/ demonstration / application	PL200 – Elevations Residential	CO 3
7-8	Creating and Editing Walkthroughs	lecture/ open discussion/ demonstration / application	PL200 Rendering and walkthrough	CO 2 CO 3
9-10	Importing and Linking Files	lecture/ open discussion/ demonstration / application	PL300 2-storey residence PL 400 PL 500	CO 1
11	Assessment of Plates	application	PLF Summative Assessment	CO 1 CO 2

Week	TOPICS	TLA	AT	COURSE OUTCOMES
				CO 3

10. Textbook:

11. Course Evaluation

Student performance will be rated based on the following:

Assessment Tasks		Weight	Minimum Average for Satisfactory Performance
CO 1	PL100	5%	75%
	PL200	10%	75%
	PL300	10%	75%
	PLF	20%	75%
CO 2	PL400	10%	75%
	PLF	10%	75%
CO 3 CO 1	PL100	5%	75%
	PL500	10%	75%
	PL600	10%	75%
	PLF	10%	75%
TOTAL		100%	75%

The final grades will correspond to the weighted average scores shown below

Average	Grade	Average	Grade
Below 60	5.00	83 – 84	2.00
75 – 76	3.00	85 – 88	1.75
77 – 79	2.75	89 – 92	1.50
79 – 80	2.50	93 – 95	1.25
81 – 83	2.25	96 – 100	1.00

o Other Course Policies

a. Attendance

According to CHED policy, total number of absences by the students should not be more than 20% of the total number of meetings or 9 hrs for a three-unit-course. Students incurring more than 9 hours of unexcused absences automatically gets a failing grade regardless of class standing.

b. Honor, Dress and Grooming Codes

All of us have been instructed on the Dress and Grooming Codes of the Institute. We have all committed to obey and sustain these codes. It will be expected in this class that each of us will honor the commitments that we have made.

For this course the Honor Code is that there will be no plagiarizing on written work and no cheating on exams. Proper citation must be given to authors whose works were used in the process of developing instructional materials and learning in this course. If a student is caught cheating on an exam, he or she will be given zero mark for the exam. If a student is caught cheating twice, the student will be referred to the Prefect of Student Affairs and be given a failing grade.

c. Consultation Schedule

12. Other References

Autodesk Revit Architecture 2013 essentials / James Vandezande, Phil Read, Eddy Krygiel.

Mastering Revit architecture 2010 / Greg Demchak, Tatjana Dzambazova, Eddy Krygiel.

Autodesk Revit architecture 2012 essentials : Autodesk official training guide / Phil Read, Eddy Krygiel, James Vandezande.

13. Course Materials Made Available